Vietnamese – German University

Faculty of Engineering

Computer Science Program

PRACTICAL TRAINING REPORT

Company:	AIVISION JOINT STOCK COMPANY
Intern period:	11/03/2026 – 10/07/2026
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Intake:	2022

2026, Viet Nam

Declaration

I hereby declare that this internship report has been created solely by myself. I affirm that I have not utilized any sources other than those specified in the references section of this report.

Furthermore, I assert that all information presented within this report is based on my own observations, experiences, and analyses during the course of my internship at AIVISION JOINT STOCK COMPANY. Any contributions or assistance provided by others have been acknowledged in the appropriate section of this report.

I understand the importance of academic integrity and take full responsibility for the originality and authenticity of this work.

Author	**Company Representative**
______________________________	______________________________
Full Name: ____________________	Full Name: ____________________
Date: ________________________	Date: ________________________

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Abstract

This report documents a four-month practical training phase as a Business Analyst Intern at AIVISION Joint Stock Company, an AI and software development firm based in Vietnam that builds and outsources intelligent applications for major technology clients. The internship took place from March 11 to July 10, 2026.

The core challenge addressed throughout the internship was bridging the communication gap between external customers and the internal development team. Operating in the role of a Business Analyst Intern, the assigned tasks spanned most of the software development lifecycle, including requirements gathering, wireframe and UI designing, project management, documentation, and quality control.

The work was carried out using an industry-standard toolset comprising Azure DevOps for project tracking and collaboration, Figma and FigJam for wireframing and mockup creation, and React with TypeScript for the detailed implementation of front-end designs. Microsoft Office and other industry-standard AI tools were also utilized for Business Analysis tasks, such as NotebookLM for generating meeting minutes and Claude/ChatGPT for daily task management and documentation.

As a result of this internship, end-to-end solutions were successfully delivered by translating customer requirements into clear, actionable specifications for the development team. Through this experience, practical competencies were developed in requirements gathering, project management, and wireframe design — skills that are essential to the role of a Business Analyst in a modern software organization.

Chapter 1

Introduction

Practical training is a compulsory component of the Computer Science program at the Vietnamese-German University (VGU), designed to provide students with first-hand exposure to professional working environments before graduation. This report documents my four-month internship placement undertaken at AIVISION Joint Stock Company from March 11 to July 10, 2026.

I came across this opportunity through a Business Analyst Intern position advertised on LinkedIn. With no prior professional experience in the role, I was drawn to the position for several practical reasons: it was a manageable four-month commitment, the office was conveniently close to home, the internship came with a reasonable allowance, and the company had a strong professional reputation. After submitting my CV, I was invited for an interview and was pleased to receive an offer shortly after.

My internship at AIVision began with an onboarding period where I assisted a senior Business Analyst on smaller-scale outsourcing software projects for clients such as GeneSolutions. This initial phase helped me build foundational skills in requirements gathering, wireframing, and documentation in a supported environment. After roughly one month, I took on greater independence and began managing three to five larger AI-driven projects simultaneously for major enterprise clients, including Masan and INNOVA US.

Going into this placement, my main personal goal was straightforward: to learn what it actually means to work as a Business Analyst — not from textbooks, but through real interactions with clients and development teams, solving real problems and delivering real solutions.

The remainder of this report is structured as follows. Chapter 2 provides a profile of AIVision as an organization. Chapter 3 outlines the key problems and challenges that I encountered during the placement. Chapter 4 describes the technologies and tools exercised throughout my internship. Chapter 5 details the tasks assigned and the methodology applied to complete them. Chapter 6 reflects on intra-organization communication and team dynamics. Chapter 7 presents the results produced and the skills developed. Finally, Chapters 8 and 9 offer my personal reflections and concluding remarks, respectively.

Chapter 2

Profile of the Organization

2.1 Company Overview and History

AIVision Joint Stock Company (aivision.vn) is a leading artificial intelligence and digital transformation enterprise headquartered in Ho Chi Minh City, Vietnam. The company was founded in January 2023 by Dr. Giang Vo, an AI scientist with fifteen years of international research and industry experience in Singapore and the United States. Established with the vision of bringing world-class AI technology to the Vietnamese market, AIVision grew at a remarkable pace, evolving into a million-dollar startup within its first year of operation.

2.2 Leadership and Workforce

AIVision maintains a highly specialised technical workforce of over 150 AI experts. The leadership and research teams are composed of elite professionals, including scientists holding PhD degrees from the National University of Singapore (NUS) and the University of Cambridge, as well as senior engineers with prior experience at major global technology firms such as NVIDIA (USA), alongside specialists from Germany and Singapore. This concentration of international talent underpins the company's capacity to develop proprietary, research-grade AI solutions at a commercially competitive pace.

2.3 Core Products and Services

AIVision develops proprietary AI models, retaining 100% intellectual property rights, optimised for the Southeast Asian market through the use of Edge Computing and real-time data synchronisation. The company offers four core enterprise solution lines:

- **AI Display Scoring (AIVision Display Tracking).** An automated Computer Vision system that evaluates retail product displays by recognising products, counting Stock Keeping Units (SKUs), and scoring Planogram compliance. The system incorporates 360-degree panorama stitching and fraud detection, achieving 97–100% accuracy with a maximum processing latency of under one second.

- **Agentic AI.** A next-generation multi-agent platform integrating Large Language Models (LLMs) such as GPT-4 and Llama. Acting as a "Digital Operations Copilot", it automates complex business workflows, customer service interactions, document summarisation, and data analysis through autonomous planning and tool use.
- **AI Camera Analytics.** A smart video analysis system integrated into existing surveillance infrastructure, designed to automatically detect retail fraud, monitor factory security, perform Automatic Number Plate Recognition (ANPR), and generate foot-traffic heatmaps.
- **Custom AI Software.** Tailored enterprise platforms including an AI HR Platform that automates 90% of the recruitment process via CV parsing and intelligent candidate matching, as well as Computer Vision applications for the healthcare sector to track and analyse dermatological treatments.

2.4 Key Achievements and Client Portfolio

AIVision's solutions have been validated in large-scale, real-world production environments. The company serves a portfolio of billion-dollar corporations and international brands, including Masan, Pepsi, Marico, VNG, and INNOVA US. Most notably, AIVision successfully deployed its AI Display Scoring platform across Masan's network of over 500,000 retail points of sale, building a system capable of handling more than 5,000 concurrent users while maintaining a 99% customer satisfaction rate.

2.5 Global Presence and Expansion Strategy

While headquartered in Ho Chi Minh City, AIVision maintains a strong international presence with physical offices in Singapore, Bangkok (Thailand), and Texas (USA). The organization is actively executing a global expansion strategy with the ambition of becoming a strategic AI partner in markets such as Australia, Mexico, and the broader United States by 2029.

Chapter 3

Problems Identified and Challenges Faced

This chapter outlines the core business problems that motivated the projects undertaken during the internship, as well as the practical challenges encountered while executing the role of a Business Analyst.

3.1 Business Problems Addressed

3.1.1 AI Display Tracking (Masan, Syngenta, P&G)

For large Fast-Moving Consumer Goods (FMCG) companies operating across massive retail networks, manually tracking product displays across thousands of points of sale is practically infeasible. Clients such as Masan, Syngenta, and P&G engaged AIVision's AI Display Scoring platform to address several interconnected problems. First, manually identifying products and counting SKUs on store shelves was slow, error-prone, and could not scale. Second, ensuring that retail outlets actually arranged products according to the brand's agreed Planogram layout was difficult to enforce and verify at scale. Third, detecting display fraud — such as misreported compliance — across a vast network was nearly impossible without an automated system. Finally, gathering intelligence on competitor product placements required human fieldwork that was both costly and inconsistent. AIVision's Computer Vision platform, incorporating 360-degree panorama stitching and automated fraud detection, was the technical solution proposed to address all of these problems simultaneously.

3.1.2 Agentic AI (INNOVA US)

INNOVA US is an American automotive technology company facing two core data challenges in the automotive diagnostics domain. First, valuable technical knowledge — spread across thousands of unstructured manufacturer documents, OEM websites, and automotive forums — could not be efficiently extracted, normalised, and made queryable at scale. Second, there was no automated system to collect and categorise the vast volume of owner-reported Q&A discussions scattered across the internet for specific vehicle models. Both problems required AI-driven pipelines to transform unstructured, dispersed data into structured, actionable datasets suitable for powering diagnostic tools.

3.1.3 Software Development (GeneSolutions)

GeneSolutions, a Vietnamese healthcare company specialising in genetic testing and prenatal care, required a comprehensive mobile application to support expectant mothers throughout their pregnancy journey and beyond. The underlying problems were numerous: pregnant women lacked access to timely, personalised medical guidance outside clinic hours; parents had difficulty tracking critical prenatal milestones and newborn vaccination schedules; financial planning for pregnancy costs was manual and opaque; and newborn development tracking was fragmented across physical records. Additionally, from a business operations perspective, the marketing team had no way to update app content or personalisation settings without requiring a full software release. The project therefore required both a patient-facing product and an internal admin portal to solve these problems holistically.

3.2 Challenges Faced During the Internship

3.2.1 Challenging Client Requirements

A recurring challenge was managing client expectations around what could realistically be delivered within the agreed timeline and budget. For example, during the Masan display tracking project, the client requested a 98% AI detection accuracy rate under a tight delivery schedule. Achieving such a level of precision with a newly deployed model in a live, uncontrolled retail environment required careful negotiation, setting realistic milestones, and managing the client's expectations transparently throughout the project.

3.2.2 Vague and Incomplete Requirements

Clients often provided requirements that lacked the specificity needed for technical implementation. Rather than concrete definitions, initial briefs were frequently high-level and ambiguous. Extracting the precise details required extensive follow-up — for instance, when designing the user interface for the GeneSolutions mobile application, it was often necessary to ask detailed questions about every screen: what elements were to be included, what each button should trigger, what the next page in a flow should look like, and so on. This process of progressive clarification was time-consuming but essential to avoid costly rework during development.

3.2.3 Communication Gap with Non-Technical Clients

Many client stakeholders did not have a technical background and were therefore unable to communicate directly with AIVision's engineering team. This created an indirect communication chain in which clients would relay requirements through their own internal

BA contacts, who would then communicate with me as the point of contact at AIVision. Each additional layer introduced the risk of information being lost, distorted, or delayed, making it critical to document every discussion carefully and confirm understanding at each step.

3.2.4 Scope Creep and Mid-Project Requirement Changes

A persistent challenge across multiple projects was scope creep — clients requesting new features or changes after the scope had already been formally agreed in the contract. While some requests were reasonable and accommodatable, others were not feasible within the existing timeline or budget. Managing these situations required clear communication about the contractual boundaries and, when appropriate, negotiating a formal scope change process rather than absorbing requests silently.

3.2.5 Expectation of Constant Availability

Several clients expected immediate responses during evenings, weekends, and outside of standard working hours. While responsiveness is important in a client-facing role, this created pressure and required setting professional boundaries while maintaining a positive working relationship.

3.3 A Specific Challenge faced: AI Detection Failure in the Masan Project

One of the most significant challenges encountered during the internship was a critical AI detection failure that emerged mid-production on the Masan display tracking project.

At the outset, the client had agreed to the operating assumptions of AIVision's detection algorithm, which was designed to work with a single row of SKUs arranged on each shelf of a display rack. Initially, the system performed as expected. However, as deployment scaled, a growing volume of complaints about low detection accuracy was received. Upon investigation, it was discovered that store staff at many locations had been stacking multiple rows of products on a single shelf — a real-world behaviour that fell outside the algorithm's designed operating conditions and caused widespread misdetection.

To resolve the issue, the following steps were taken. First, the problem was diagnosed and documented in detail, including the specific request numbers, timestamps, incorrectly detected SKUs, and recorded accuracy figures. Second, the situation was escalated internally to the development team, with a clear explanation of the root cause and a proposed solution involving a different detection algorithm better suited to variable stacking patterns. Third, a meeting was arranged with the Masan team to transparently communicate the problem, explain the cause, and present the proposed fix. Ultimately, the client was receptive to the solution, and following their agreement, the updated model was deployed. Detection accuracy returned to an acceptable level and complaints subsided.

This experience highlighted the importance of proactive monitoring, clear internal escalation, and transparent client communication when production issues arise.

Figure 3.1: AI mis-detection caused by double-row SKU stacking on a Masan retail display.

As illustrated in Figure 3.1, the Kokomi and Omachi instant noodle packages were stacked in two rows on a single shelf rather than one. Since the detection algorithm was designed under the assumption of a single product row per shelf, products were miscounted and Planogram compliance scores were reported inaccurately. This real-world stacking behaviour, which had not been accounted for in the original deployment agreement, was the root cause of the widespread accuracy complaints received from Masan's field network.

Chapter 4

Technologies, Instruments, and Programming Frameworks

This chapter describes the tools, frameworks, and platforms exercised throughout the internship, grouped by their primary purpose.

4.1 Wireframing and UI Design

4.1.1 Figma and FigJam

Figma is a browser-based, collaborative interface design tool widely used in the software industry for wireframing, prototyping, and high-fidelity UI design. FigJam is Figma's companion whiteboarding tool, designed for brainstorming, flowcharting, and real-time team collaboration. Both tools were used extensively during the GeneSolutions mobile application project. Figma was the primary workspace for constructing wireframes, designing screen mockups, and defining button flows and navigation logic. FigJam was used for mapping out user flows and collaborating with frontend and backend developers to align on interface logic before implementation. Completed designs were also presented to the client directly from Figma during review and demo sessions.

4.1.2 Stitch AI

Stitch is an AI-powered UI design tool developed by Google that generates interface designs from natural language prompts. Rather than starting from a blank canvas, a textual description of a screen's purpose and content can be provided, and Stitch produces a first-draft UI layout automatically. During the GeneSolutions project, Stitch was used as a starting point to accelerate the early design phase — prompting it to generate initial screen drafts, which were then imported into Figma for refinement, adjustment to the client's brand guidelines, and further iteration. This workflow significantly reduced the time spent on initial layout construction.

4.1.3 React and TypeScript

React is a widely adopted JavaScript library for building component-based user interfaces, and TypeScript is a statically typed superset of JavaScript that improves code reliability and developer tooling. In addition to the Figma-based design workflow, React and TypeScript were used to vibe-code UI screens for the GeneSolutions mobile application. This approach — using an AI-assisted coding workflow to generate interface code from design intent — offered a faster alternative to manual Figma design for certain screens, allowing UI to be prototyped and iterated at a higher pace without being bottlenecked by pixel-level design work.

4.2 Project Management

Azure DevOps is a Microsoft platform that provides an integrated suite of tools for software project planning, version control, and delivery management. The primary feature used during the internship was the Boards module, which supports Agile sprint-based project management. This included creating and managing sprint cycles, defining and assigning tasks and subtasks to team members, setting priorities and deadlines, and tracking the status of each work item (e.g. To Do, In Progress, Done) across multiple projects simultaneously. Azure DevOps Boards provided a shared, real-time view of project progress for both the internal team and, where applicable, the client's project contacts.

4.3 Requirements Gathering and Documentation

Microsoft Office — primarily Word and Excel — was used for formal documentation throughout the internship. Word was used to produce technical requirements specifications, functional guidelines, and meeting minutes following client and internal team discussions. Excel served as the primary tool for structured data management, covering a wide range of tasks including bug report tracking, SKU tracking sheets, client question lists, logging detected accuracy issues, and maintaining reference lists for quality control purposes.

4.4 Quality Control

Label Studio is an open-source data labelling platform used to annotate training and evaluation datasets for AI and machine learning models. During the internship, it was used as part of the monthly quality control process for the Masan AI Display Tracking project. Specifically, Label Studio was used to review and correct incorrect bounding box annotations — cases where the AI model had mis-detected object boundaries or incorrectly identified SKU rows on retail display shelves. The corrected annotations were then compiled into a QC report and submitted to Masan for monthly tracking and reporting. The development team also leveraged this report to support model improvement.

4.5 AI Tools

4.5.1 NotebookLM

NotebookLM is an AI research and note-taking tool developed by Google that allows users to upload documents and query them conversationally. It was used during requirements gathering to analyse large volumes of client-provided documentation — extracting relevant functional requirements, summarising key points, and identifying gaps or ambiguities. NotebookLM was also used to process audio recordings of client meetings and generate structured meeting minutes, significantly reducing the time required for post-meeting documentation.

4.5.2 Claude

Claude is a large language model developed by Anthropic, used during the internship as an AI writing assistant for technical documentation. It was particularly useful for drafting and refining client-facing documents, including functional requirement guidelines, question lists sent to clients during the requirements gathering phase, SKU tracking tables, and bug report summaries. Claude's ability to maintain context across long documents made it well-suited to producing consistent, professional-quality technical writing.

4.5.3 ChatGPT and Gemini

ChatGPT (developed by OpenAI) and Gemini (developed by Google) were used as general-purpose AI assistants throughout the internship for day-to-day tasks such as drafting emails, looking up technical concepts, and supporting quick analysis or summarisation tasks. While their usage was less specialised than NotebookLM or Claude, they served as versatile productivity tools that reduced the time spent on routine cognitive tasks.

Chapter 5

Tasks Assigned

5.1 Wireframing and UI Design

A significant portion of the internship was dedicated to UI/UX design for the GeneSolutions pregnancy mobile application. The project required designing the interfaces for three distinct features: a Baby Naming tool, an After Birth tracking suite, and an internal Admin Portal. All three were delivered through a consistent iterative workflow — translating written requirements into screen designs, gathering client feedback, and refining until a prototype was ready for demo.

5.1.1 Design Workflow

The design process for each screen followed a consistent three-stage pipeline. First, the functional requirements for the screen were reviewed and distilled into a structured prompt, which was used either with Google Stitch AI or directly in VS Code using React and TypeScript via an AI-assisted vibe-coding approach to generate a first-draft layout. This initial draft served as a low-effort starting point rather than a blank canvas, allowing rapid exploration of layout options. Second, the generated screens were imported and arranged on a Figma or FigJam board, where navigation flows, button logic, and screen transitions were mapped out and refined. Third, the resulting prototype was reviewed by the client in sessions held approximately two to three times per week. Early review cycles focused primarily on high-level flow and structure, while later rounds narrowed to finer details such as label wording, component spacing, and interaction states. This iterative loop continued for approximately three weeks until a stable prototype was approved for handoff to the development team.

5.1.2 Baby Naming Tool

The Baby Naming tool is a personalised name recommendation feature that combines Numerology and Feng Shui to help expecting mothers find a meaningful name for their child. The designed screens covered the full user flow across six stages.

The entry screen allows mothers to select their preferred naming method — Numerology (based on birth date) or Feng Shui (based on birth year and elemental destiny). A detailed input form then collects the baby's gender, the family surname, the mother's preferred naming style, the expected or actual birth date, and a dropdown of blessings or character traits the mother wishes for her child.

Following submission, the app generates a curated list of name suggestions with a brief summary visible inline and a full analytical breakdown accessible on tap. A history log screen allows mothers to revisit previously generated lists. The deep analytics view presents a four-part breakdown of the selected name, covering the linguistic and emotional story behind the name, a Pythagoras birth chart decoding, an integrated name-and-birth-date compatibility grid, and personalised parenting insights derived from the child's numerology profile. Finally, a shareable "Certificate of Love" screen was designed to allow mothers to export and share a visual keepsake of the baby's name and its meaning to social platforms. The complete screen flow is illustrated in Figure 5.1.

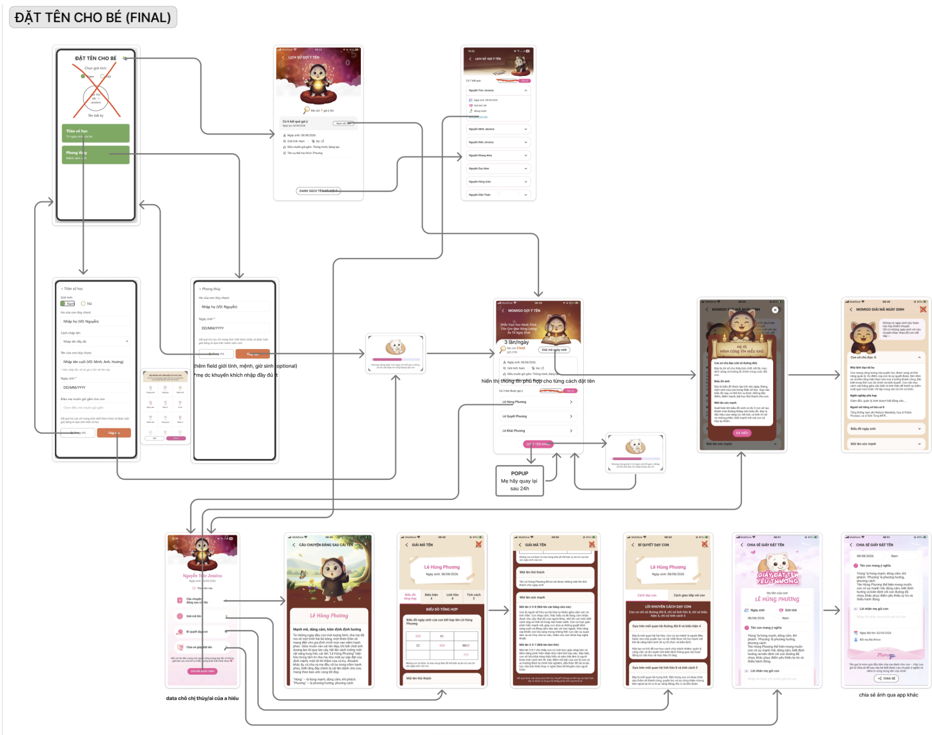

Figure 5.1: FigJam flow diagram for the Baby Naming tool. Click here to view the interactive FigJam board.

5.1.3 After Birth Suite

The After Birth suite activates dynamically once a mother records that she has given birth. At 41 weeks of pregnancy, a pop-up prompts the mother to confirm delivery, which triggers a transition from prenatal tracking mode into a postpartum dashboard personalised to the newborn.

The onboarding flow guides the mother through setting up the baby's profile — uploading a photo, entering the full name, gender, birth date and time, and initial weight and height measurements. The central dashboard provides a real-time health summary showing the baby's exact age in months and days, the next upcoming vaccination with a countdown, and the latest growth snapshot with delta indicators from the previous entry. A multi-child management screen was also designed to allow mothers to switch between profiles for multiple children.

The Growth Tracking module features interactive weight and height charts plotted over a monthly timeline, with WHO growth standard benchmarks overlaid to automatically classify the baby's development as within the normal range or flagged as below average. An entry form allows new measurements to be logged by date, with an auto-generated quick analysis summary.

The Immunisation Scheduler organises all required vaccines month by month, explicitly naming each critical shot — including Hepatitis B, the 5-in-1 Quinvaxem/Pentaxim, Rotavirus, Pneumococcal, etc. — to ensure no doses are missed. Finally, the Milestone Journal provides a visual timeline feed where mothers can log dated entries with photos and videos, categorised by milestone type, health events, or sleep records. The full postpartum flow is illustrated in Figure 5.2.

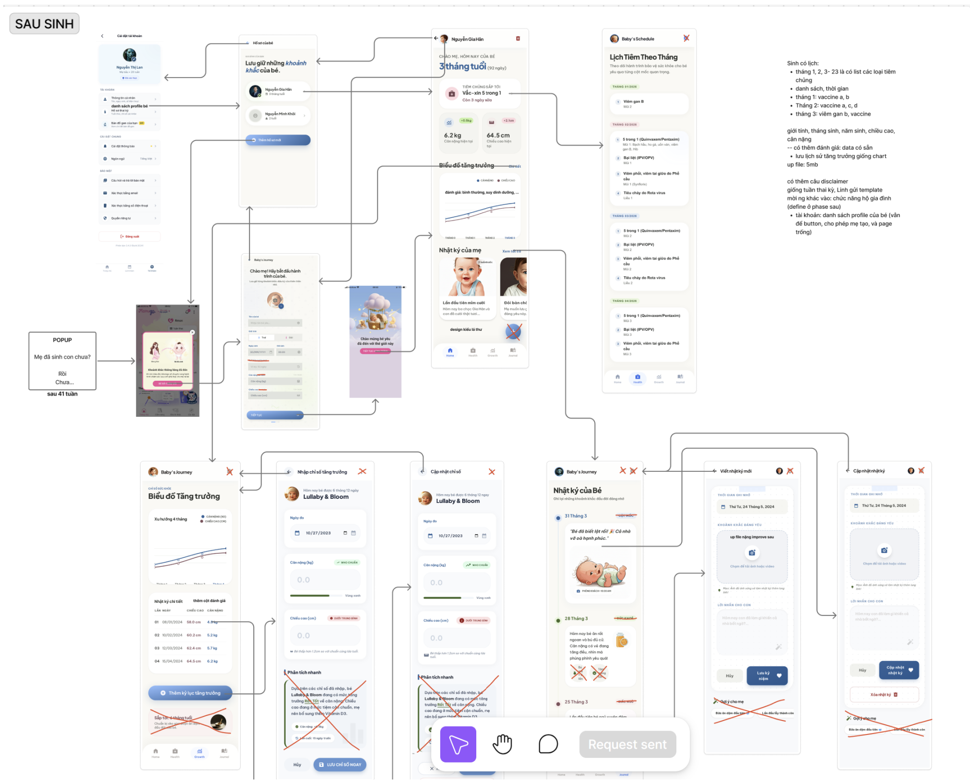

Figure 5.2: FigJam flow diagram for the After Birth suite. Click here to view the interactive FigJam board.

5.1.4 Admin Portal

The Admin Portal is a web-based Content Management System (CMS) designed for use by the GeneSolutions operations and marketing team. It provides centralised control over all content, media, clinical surveys, and user accounts within the mobile application — removing the dependency on software releases for routine content updates.

The portal's authentication screen provides a secure login entry point restricted to credentialled GeneSolutions staff. Upon login, a welcome dashboard greets the administrator and defers detailed analytics to a future release phase.

The Handbook Management module supports two content pipelines: native article creation, where admins write and publish articles from scratch using a rich text editor with category and author assignment; and Sale Portal fetching, which pulls pre-vetted corporate content from a linked business platform. Fetched articles have their core fields permanently locked to protect brand and medical accuracy, with admins only permitted to configure app-specific metadata such as categories and target pregnancy stage tags.

The Media Library module covers three sub-sections: banner management with drag-and-drop reordering, a general photo asset repository, and a theming dashboard for applying visual templates across the application.

The Clinical Checklist module provides a form builder interface for creating and versioning clinical surveys, with per-step and per-field configuration including field type, validation rules, and optional or mandatory toggles.

Finally, the User and Admin Access Control module maintains separate registries for admin accounts and end-user accounts, with a detailed per-user view exposing account metadata and synchronised health parameters such as weight, height, blood type, and current pregnancy week. The portal flow is illustrated in Figure 5.3.

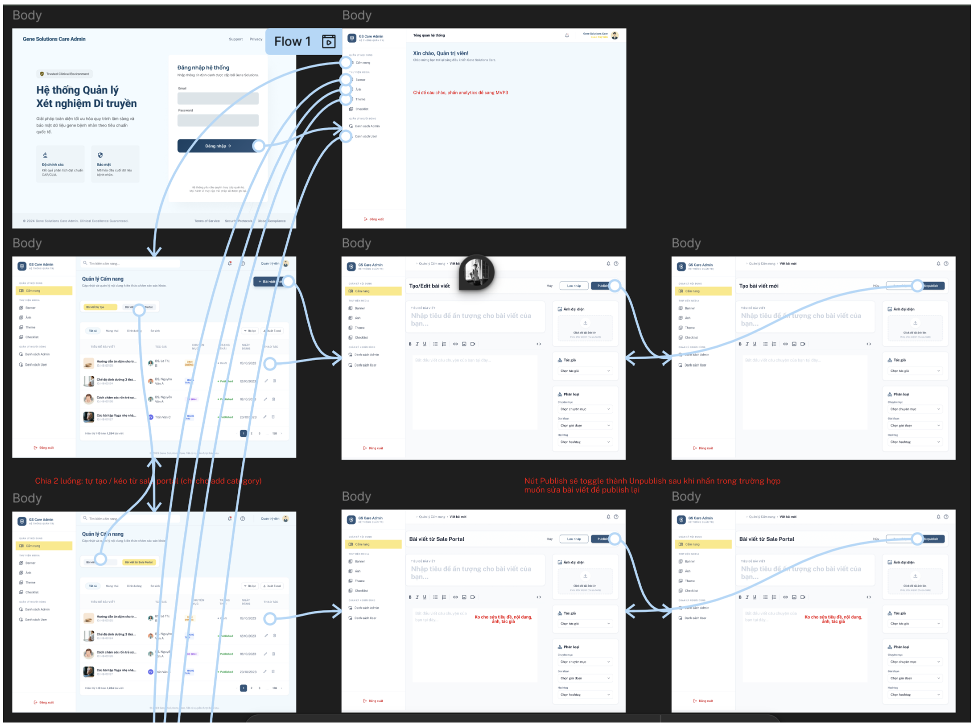

Figure 5.3: FigJam flow diagram for the GS Care Admin Portal. Click here to view the interactive Figma wireframe.

5.2 Project Management

Project management was a core responsibility throughout the internship, spanning five concurrent projects: Masan, Syngenta, P&G, INNOVA US, and GeneSolutions. The primary tools were Azure DevOps Boards for sprint planning and task tracking, Microsoft Teams for meeting coordination, and Microsoft Excel for supplementary tracking of timelines, deadlines, and ownership.

5.2.1 Sprint Planning on Azure DevOps

Each project was managed through one-week sprints on Azure DevOps Boards. At the start of each sprint, tasks were created and broken down into smaller subtasks, each assigned a deadline, a person in charge, and a status label. A typical sprint board carried five to six high-level tasks spanning all roles — Business Analysis, Quality Control, and multiple development tracks — with each task decomposed into several subtasks to reflect the actual unit of daily work. Across the five active projects, this amounted to coordinating task assignment and progress tracking for a team of approximately 15 people, including 8 developers. Figure 5.4 shows a sample sprint board for the Masan project.

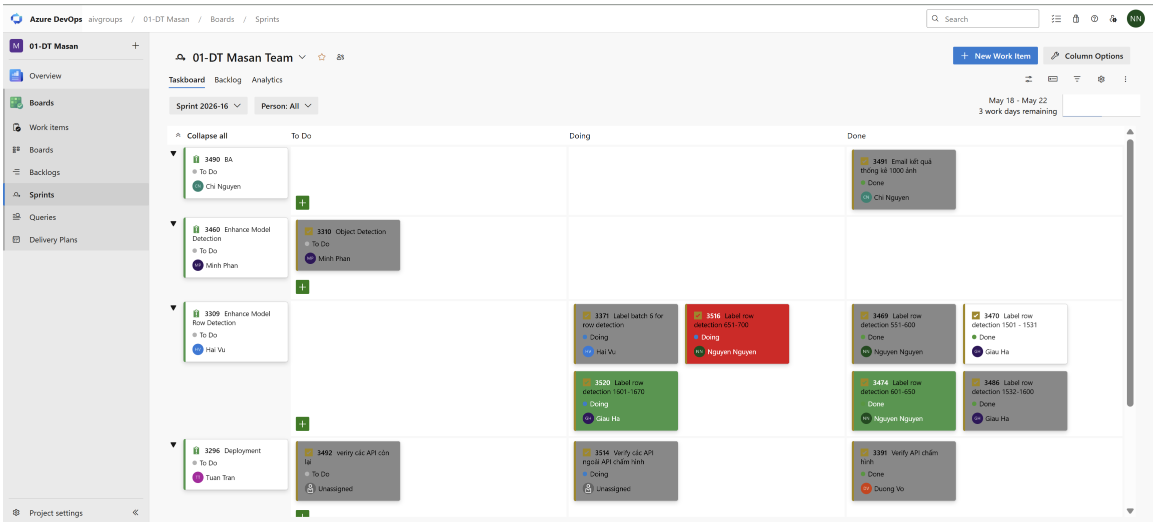

Figure 5.4: Azure DevOps sprint board for the Masan project (Sprint 2026-16).

5.2.2 Meeting Schedule and Facilitation

Internal meetings followed a fixed weekly structure designed to bookend each work week. A Monday morning session at 10:00 was held at the start of each week to align on sprint goals, assign priorities, and review any incoming requirements or client feedback from the previous week. A Friday afternoon session at 16:30 closed the week with a sprint review — going over completed tasks, updating timelines, and flagging any blockers or scope changes for the following sprint.

In addition to these weekly anchors, daily stand-up meetings were held twice a day at 08:00 and 16:45 on all other weekdays. The morning stand-up opened the day by surfacing blockers and confirming each team member's focus, while the afternoon stand-up provided an end-of-day sync to capture progress, note any issues, and adjust task statuses on the board before the next morning. All meetings were conducted over Microsoft Teams and facilitated internally.

5.2.3 Task and Timeline Tracking

Beyond the Azure DevOps boards, a supplementary Excel tracker was maintained to provide a higher-level view of timelines and ownership across all projects simultaneously. The tracker consolidated key milestones, delivery deadlines, and responsible parties in a format suited to cross-project visibility — particularly useful when reporting status to senior management or when a client requested an overview of the current delivery schedule. An example of the weekly tracking sheet is shown in Figure 5.5.

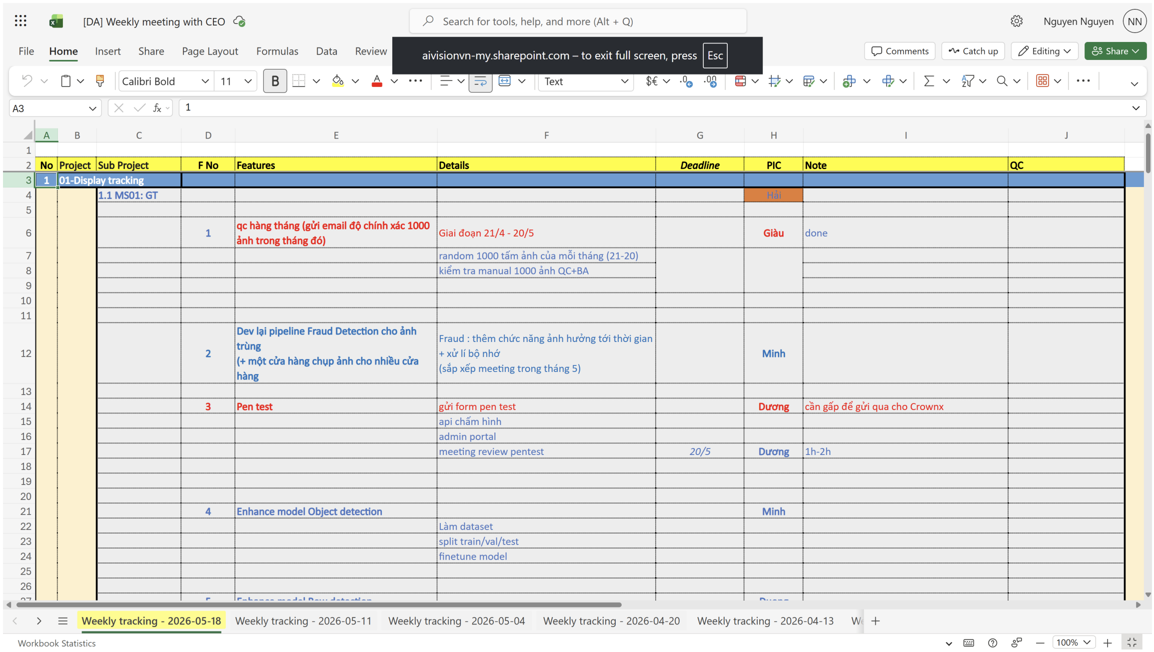

Figure 5.5: Weekly task tracking spreadsheet (week of 18 May 2026).

5.2.4 Client Feedback and Dev Team Coordination

A significant part of the project management role involved acting as the interface between client feedback and the internal development team. When clients raised issues, change requests, or new requirements in meetings or over email, these were captured, clarified, and translated into actionable tickets on the Azure DevOps board with appropriate priority labels, acceptance criteria, and assigned developers. This ensured that nothing raised in client communication was lost in translation, and that developers received clear, structured work items rather than raw, unfiltered feedback.

5.3 Requirements Gathering and Documentation

Requirements gathering and documentation work spanned all five projects and produced a range of deliverable types — structured requirement specifications, issue trackers, quick-turnaround question lists, field operation guidelines, and meeting minutes. Microsoft Word, Excel, and PowerPoint each served a distinct purpose depending on the nature of the output.

5.3.1 Requirements Specifications

The INNOVA US project involved two primary requirements: building a structured database of vehicle symptoms from Technical Service Bulletins (TSBs), and crawling question-and-answer data organised by Year, Make, and Model. The client supplied original requirement documents in PDF format. These were reviewed, then converted into Excel spreadsheets to impose a consistent tabular structure — separating each requirement into discrete fields covering scope, input/output format, acceptance criteria, and technical notes. The structured format made it significantly easier to distribute the requirements to the development team in an actionable form and to cross-reference individual items during review. Following an internal walkthrough, the requirements were presented back to the client in review meetings to confirm interpretation and obtain approval before development began. The structured requirements sheet for R1 is shown in Figure 5.6.

R1 – Structured Symptoms Database

Mục tiêu: Tự động hóa việc xây dựng Symptoms Database toàn diện bằng cách xử lý unstructured data (TSB, OEM, forum) thông qua AI → structured data phục vụ chẩn đoán & ước tính sửa chữa

#	Feature	Sub-feature	Description	Input	Output	Note
1	Tổng quan hệ thống	Mục tiêu dự án	Xây dựng Symptoms Database từ các nguồn kỹ thuật không có cấu trúc (file PDF TSB, website OEM, forum ô tô) sử dụng AI/NLP để trích xuất dữ liệu chẩn đoán & sửa chữa có cấu trúc.	*TSB PDF, website OEM, dữ liệu forum*	*Symptoms Database có cấu trúc*	Đầu ra chính: liên kết Xe + DTC + Triệu chứng + Phương án sửa
2		Kiến trúc hệ thống	Pipeline xử lý 2 giai đoạn: • Giai đoạn 1 – Xử lý tài liệu TSB (độ chính xác cao, nguồn có cấu trúc) • Giai đoạn 2 – Xử lý website OEM & forum internet (phạm vi rộng hơn)	*Tất cả nguồn kỹ thuật có sẵn*	*Database được điền đầy đủ theo cả 2 giai đoạn*	Giai đoạn 1 ưu tiên trước (dữ liệu đáng tin cậy nhất)
3		Cấu trúc Database	Mỗi bản ghi liên kết: Xe (YMME) ↔ DTC ↔ Triệu chứng ↔ Phương án sửa. • Phase 1 TSB có DTC: đủ 4 trường • Phase 1 TSB không DTC: YMME + Triệu chứng + Sửa (không có DTC) • Phase 2 OEM/Forum: đủ các trường, DTC là tùy chọn	—	*Các bản ghi quan hệ được liên kết*	YMME = Year (Năm), Make (Hãng), Model (Dòng xe), Engine (Động cơ)
4		Yêu cầu chuẩn hóa	Tất cả giá trị trích xuất phải được chuẩn hóa theo danh sách chuẩn của Innova: • Xe: POLK_YMME_v1.xlsx • Triệu chứng: Danh sách thuật ngữ triệu chứng • Tên sửa chữa: INNOVA_FIXNAME_v1.xlsx • Tên phụ tùng: INNOVA_PARTNAME_v1.xlsx	*Giá trị thô do AI trích xuất*	*Bản ghi đã chuẩn hóa, nhất quán*	Nếu không khớp: ghi nhãn 'Cannot match'
PHASE 1 – XỬ LÝ TÀI LIỆU TSB						
Bước 1 ‣ Xác định TSB cần xử lý						

Figure 5.6: Structured Excel requirements sheet for INNOVA US. Click here to view the full document.

5.3.2 SKU and Bug Tracking

Excel spreadsheets were used extensively across three projects — Masan, Syngenta, and P&G — to maintain structured records of SKU data used for model training and to log detection issues reported in the field.

For model training purposes, a dedicated SKU tracking sheet was maintained for each project. Each row in the sheet represented a single SKU and recorded the SKU identifier, product name, image, received date, status, etc. These sheets served as the ground truth reference for the development and QC teams when preparing and validating training datasets. An example of the P&G SKU tracking sheet is shown in Figure 5.7.

For the Masan project, a separate bug tracking sheet was maintained alongside the SKU registry. Masan's sales staff used a mobile application portal to photograph retail shelves and receive AI detection results in real time. When a detection failed or produced incorrect output, staff would report the issue directly via a dedicated group chat. These reports were captured and logged with each row recording the message link, request number, date

	BRAND	BARCODE	PRODUCT NAME		Link	Hình ảnh
SKU003	PANTENE	4902430754583	DG PTN NB CSHUTON CSAU 450MLX9	DG_PTN_NB_CSHUTON_CSAU_450MLX9	https://drive.google.com/uc?export=view&id=1Fl0llKlRRDl23T74uW1n5ynT4LjbrMbq	
SKU005	HERBAL ESSENCES	8001090222572	DX HE DAUTAY & BACHA 400MLX6	DX_HE_DAUTAY_&_BACHA_400MLX6	https://drive.google.com/uc?export=view&id=1gEaepmHj1hVRtMlPX3YA_z2gkcxlP34L	
SKU007	PANTENE	4902430754590	DX PTN NB CSHUTON CSAU 400GX9	DX_PTN_NB_CSHUTON_CSAU_400GX9	https://drive.google.com/uc?export=view&id=1_m41GUtHGG43dHBX5ORxZ4ePxYgy3FIO	

Figure 5.7: SKU tracking spreadsheet for the P&G project.

of incident, the customer's complaint description, and proof photographs. This tracker served both as a running log of all reported issues and as the primary input for the monthly Quality Control review cycle, where accumulated bugs were batched and submitted to the development team for model retraining.

5.3.3 Async Question Lists

Not all clarifications warranted a formal meeting. When the development team had a set of discrete questions for a client — such as ambiguous field definitions, edge case handling, or data format confirmations — an asynchronous question list was prepared in Excel. The template allowed developers to enter their questions alongside their current understanding or proposed interpretation, which the client could then fill in at their own pace and return. The completed sheet was reviewed and summarised before being fed back to the development team. This approach reduced the overhead of scheduling short alignment meetings while keeping client communication traceable and documented.

For the INNOVA US project, question lists were used extensively during the TSB processing phase to resolve ambiguous fix-name mappings. Each row in the sheet identified the source file or field, the raw value extracted by the AI, the standardised value the model had mapped it to, and an open question asking the client to confirm or correct the mapping. For example, one entry queried whether the raw fix name "replacing the URL(s)" should map to the standard "Cannot match fixname" given the absence of a close match in the INNOVA fix-name list, while another asked whether all 2024 model variants containing the suffix "e" should be treated uniformly as electric vehicles. Client responses were returned in a dedicated answer column, providing clear decisions that could be applied consistently across the dataset. An example of the INNOVA question list is shown in Figure 5.8.

File / Trường	Câu hỏi	Câu trả lời
T-SB-0030-25	Fixname raw: Install Jumper Harness Fixname std: Repair Brake Light Switch Wiring Dựa vào đâu để map ra fix name trên?	Install Jumper Harness => Repair Wiring Trong TSB có nhắc đến nội dung: C1425 (Open in Stop Switch Circuit) => Brake Light Switch Wiring
08-032-25	model raw: 500e model std: 500 với những case như này thì map hết những cái liên quan tới 500 hay sao? fixname raw : replacing the URL(s) fixname std: Cannot match fixname Dựa vào đâu để map ra được Cannot match fixname?	- Trong case này, năm 2024 chỉ có 1 model là 500 (có 3 trim nhưng cả 3 đều có engine là electric). Ký tự "e" trong trường hợp này được hiểu là electric, như vậy tất cả model 500 năm 2024 đều phù hợp với raw data. - fix name raw: replacing the DRL(s). DRL = Daytime Running Light. Replacing the DRL(s) được hiểu là thay thế Daytime Running Light, bao gồm cả cụm (chóa đèn, bóng đèn bên trong,...) Trong INNOVA FIXNAME LIST có fix name: Replace Daytime Running Light Bulb, chỉ là thay thế bóng đèn bên trong, chưa hoàn toàn phù hợp -> để Cannot match fixname
6661950	fixname raw : Repair Chafed Wire Harness fixname std: Repair Hybrid/EV Drive Motor Wiring Dựa vào đâu để map ra fixname trên?	Chevrolet Blazer EV => xe điện, dựa vào hình ảnh do TSB đưa ra + kĩ thuật ô tô => chọn fix Repair Hybrid/EV Drive Motor Wiring
23-047-24	fix name raw: torquing hood hinge-to-hood fasteners fix name std: Inspect Hood Latch, Hinges, and Fastening Bolts Dựa vào đâu để map ra fixname trên?	Fix name raw và Fix name std có ý nghĩa tương đồng.

Figure 5.8: Async question list for the INNOVA US project. Click here to view the full document.

5.3.4 Field Photography Guidelines

For the Syngenta project, the AI display tracking system required a consistent supply of planogram photographs taken by Syngenta's sales staff across retail stores. To ensure photo quality was sufficient for accurate SKU detection, a set of photography guidelines was produced as a PowerPoint presentation. The document covered camera angle, recommended distance from the shelf, lighting conditions to avoid glare or shadow, phone or camera hold technique, resolution requirements, and step-by-step instructions for capturing multi-section displays. The guidelines were developed in collaboration with Syngenta's own Business Analyst, whose prior experience with similar field operations informed much of the practical advice. The final document was distributed to the Syngenta sales team for use during their store visits. Click here to view the photography guidelines.

5.3.5 Meeting Minutes

Following client-facing and key internal meetings, meeting minutes were written up in Microsoft Word. Each document captured the agenda, key discussion points, decisions reached, and action items assigned with responsible parties and target dates. Where audio recordings were available, NotebookLM was used to assist with transcription and extraction of key points before the minutes were drafted and circulated. Minutes served both as a reference record for the team and as a formal confirmation of agreed scope or decisions that could be revisited in later sprints. An example meeting minute document for the GeneSolutions mobile application project — covering date and time, discussion points, agreed decisions, timeline, and next steps — can be accessed here.

5.4 Quality Control

Quality control work during the internship was concentrated on the Masan AI Display Tracking project, where a monthly QC cycle was established to measure detection accuracy, surface systematic errors, and provide the development team with annotated data for model improvement.

5.4.1 Monthly Sampling and Labelling

Each month, a random sample of 1,000 shelf images captured by Masan's sales staff was drawn from the detection pipeline for manual review. The sample was selected to cover a representative spread of store locations, product categories, and shooting conditions. The QC team — consisting of me and dedicated QC staff — then worked through the sampled images over a two-week window using Label Studio, an open source data labeling and AI evaluation platform . For each image, the raw bounding box detections produced by the AI model were reviewed side by side with the original photograph. Where detections were incorrect — whether due to misidentified SKUs, missed items, duplicate boxes, or mis-labelled product codes — the annotations were corrected manually by redrawing or relabelling the affected regions. An example of the Label Studio labelling interface for a Masan refrigerated display is shown in Figure 5.9, illustrating the bounding box overlay and per-region SKU assignment panel.

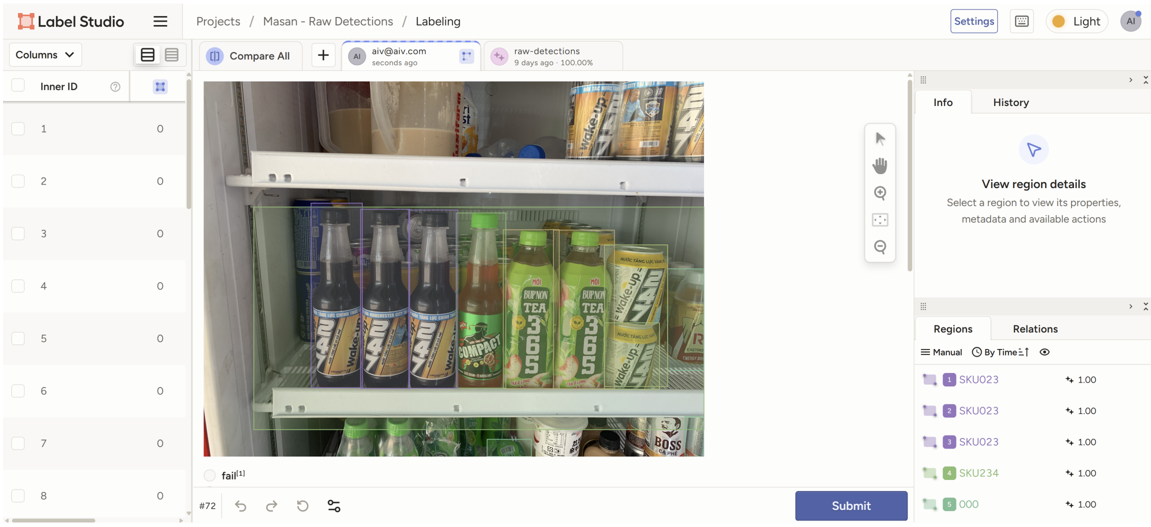

Figure 5.9: Label Studio annotation interface for the Masan project.

5.4.2 QC Report and Handoff

Upon completion of the two-week labelling window, the corrected annotations and detection outcomes were compiled into a monthly QC Excel report. The report documented per-image accuracy statistics including the number of AI-detected boxes versus ground truth boxes, True Positives (TP), False Positives (FP), False Negatives (FN), Precision, Recall, F1 Score, and Jaccard Index. For each image where corrections were made, a change log entry described the specific modification — for example, relabelling a box from `[SKU006]` to `[SKU232]`, deleting a spurious detection, or adjusting a bounding box position with its IoU delta. This report served two purposes: it was sent to Masan as the official monthly quality assurance deliverable, providing the client with a transparent record of detection performance; and a copy was shared internally with the development team to inform the next retraining cycle, giving the AI engineers a structured set of confirmed errors and corrected labels to incorporate into updated training data. The QC report is shown in Figure 5.10.

Task	Request ID	AI Boxes	GT Boxes	TP	FP	FN	Precision	Recall	F1 Score	Jaccard Index	Thay đổi so với AI (reviewer)
301	919746	10	10	10	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
302	781782	5	5	5	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
303	928348	48	49	46	2	3	95.8%	93.9%	94.8%	90.2%	Sửa nhãn [SKU006] → [SKU232] Sửa nhãn [SKU006] → [SKU232] Thêm box [SKU136] tại (48.0%, 26.5%)
304	770152	26	26	26	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
305	815124	12	12	12	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
306	850469	33	31	31	2	0	93.9%	100.0%	96.9%	93.9%	Sửa vị trí [SKU116] (IoU=0.64, Δmax=7.5%) Xóa box [000] tại (61.1%, 6.7%) Xóa box [SKU218] tại (84.3%, 8.1%)
307	780227	33	33	33	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
308	677791	44	45	42	2	3	95.5%	93.3%	94.4%	89.4%	Sửa nhãn [000] → [SKU146] Sửa nhãn [SKU027] → [SKU211] Thêm box [SKU027] tại (70.1%, 46.4%)
309	804889	41	42	41	0	1	100.0%	97.6%	98.8%	97.6%	Sửa vị trí [SKU201] (IoU=0.66, Δmax=4.7%) Thêm box [SKU201] tại (14.9%, 69.2%)
310	711871	67	67	66	1	1	98.5%	98.5%	98.5%	97.1%	Sửa nhãn [000] → [SKU023]
311	704376	27	27	27	0	0	100.0%	100.0%	100.0%	100.0%	Sửa vị trí [SKU150] (IoU=0.58, Δmax=3.5%)
312	800863	22	22	22	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
313	756166	4	4	4	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
314	725769	46	45	45	1	0	97.8%	100.0%	98.9%	97.8%	Xóa box [SKU003] tại (38.0%, 62.5%)
315	816507	51	49	49	2	0	96.1%	100.0%	98.0%	96.1%	Xóa box [SKU219] tại (60.0%, 95.3%) Xóa box [SKU219] tại (69.6%, 95.1%)
316	867682	9	9	9	0	0	100.0%	100.0%	100.0%	100.0%	✓ Không thay đổi
317	907885	27	26	26	1	0	96.3%	100.0%	98.1%	96.3%	Xóa box [SKU246] tại (2.8%, 52.0%)
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	Xóa box [SKU141] tại (97.5%, 32.9%)

Figure 5.10: Monthly QC Excel report for the Masan project. Click here to view the full report.

Chapter 6

Intra-Organization Communication

Effective communication was central to the day-to-day operations of the internship. As a Business Analyst Intern at AIVISION, I operated within a structured organizational hierarchy and relied on a combination of digital tools to stay aligned with team members across different functions and locations. This chapter describes the reporting structure, the communication tools in use, and the day-to-day practices that shaped how information was shared within the organization.

6.1 Organizational Structure and Reporting Lines

AIVISION operates with a relatively flat internal structure. At the top of the hierarchy is the Chief Executive Officer (CEO), who holds overall strategic authority and distributes high-level responsibilities to team leads. These leads — including the Development Lead, QC Lead, and BA Lead — each manage their respective teams and report back to the CEO. Interns and junior members work under these leads and receive their tasks through this chain.

In practice, my primary point of contact was the Senior Business Analyst, who assigned tasks, reviewed deliverables, and provided feedback on my work. I also interacted regularly with members of the development team and the QC team, and occasionally directly with the Development Lead and QC Lead when a discussion required their technical input or approval. Direct communication with the CEO occurred mainly in scheduled weekly meetings and occasionally in daily stand-up meetings, though these interactions were less frequent.

6.2 Communication Tools

The team used a combination of tools, each serving a distinct purpose depending on the nature of the communication.

Microsoft Teams was the primary platform for structured meetings, both internally and with clients. Video calls, screen sharing, and recorded sessions were all conducted through Teams. It served as the official meeting space for sprint planning, client requirement sessions, and cross-team syncs.

Zalo functioned as the informal, high-frequency communication layer. Group chats on Zalo were used for quick messages, urgent questions, status updates, and sharing documents such as reports and shared files. Given that Zalo is widely adopted in Vietnam, it was the preferred channel for fast internal coordination that did not warrant a formal email.

Microsoft Outlook was reserved for official and formal correspondence. Emails were used when communicating with clients in a professional capacity, sending structured deliverables, or following up on decisions made in meetings.

Viber was used specifically for communication with the Masan client team, as that was their preferred messaging platform. This required maintaining a separate communication thread for Masan-related updates and queries.

OneDrive and SharePoint served as the shared document repository. Files including requirements specifications, QC reports, meeting minutes, and tracking sheets were stored and shared through these platforms, ensuring that all team members had access to the latest versions of working documents.

6.3 Day-to-Day Communication Practices

On a typical working day, communication flowed through both synchronous and asynchronous channels. Asynchronous updates — such as sharing a completed document, flagging a blocker, or sending a question list to the development team — were handled primarily through Zalo and email, depending on the formality of the message. For example, when I needed to relay a batch of questions to the clients, these were compiled into an Excel document and shared via Zalo for the clients to respond to on their own schedule.

Internally, the team used a mix of Vietnamese and English depending on context. Vietnamese was the default language for casual conversations and quick internal exchanges, while English was used for client-facing documents, formal reports, and meetings with international or English-speaking clients.

A notable communication challenge arose from the fact that some developers worked remotely. Coordinating with remote team members required more deliberate async communication — ensuring that questions were clearly written and self-contained rather than relying on real-time clarification. Additionally, technical discussions with developers occasionally involved domain-specific terminology beyond my background as a BA intern. In such cases, I relied on the Senior BA or the relevant lead to mediate or clarify, and made a habit of documenting any new technical terms encountered during these exchanges.

6.4 Internal Meetings

AIVISION maintained a structured recurring meeting schedule. Two daily stand-ups were held via Microsoft Teams: a morning session from 8:00 to 8:15 AM where each member stated their plan for the day, and an evening session from 4:45 to 5:00 PM for progress updates and flagging blockers. Both were facilitated by the BA team, with each speaker given a maximum of two minutes. A weekly delivery meeting was held every Monday from 10:00 to 11:00 AM, combining sprint review and sprint planning for the BA team, Tech Lead, and QC team. I was responsible for preparing the BA team's project status summaries and co-presenting during this session. Additionally, a monthly KPI review was held on the first of each month, where each team member individually presented their personal KPI sheet to the CEO or line manager for performance evaluation.

Chapter 7

Results Produced and Skills Developed

7.1 Results Produced

Over the course of the four-month internship, a range of concrete deliverables were produced across the four main task areas described in Chapter 5.

UI Wireframes. Three complete UI wireframe sets were designed for the GeneSolutions mobile application: the Baby Naming Tool, the After Birth Suite, and the Admin Portal. Each consisted of approximately 10 to 15 screens, covering the full user flow from entry points through to core interactions and edge cases. All three were prototyped using Figma, iteratively reviewed against client feedback, and formally demonstrated to the GeneSolutions client team. The wireframes were approved and incorporated into the product roadmap for development beyond the internship period.

Requirements and Documentation. Approximately 10 requirements documents were produced across the active projects, including tabular requirements specifications, async question lists for the development team, SKU tracking sheets and field photography guidelines. These documents were regularly reviewed and confirmed in client meetings before being handed off to the development team as actionable inputs.

Sprint and Project Management. Approximately 30+ weekly sprint boards were maintained on Azure DevOps over a two-month active sprint period, covering all ongoing projects simultaneously. Task tracking, client feedback routing, and cross-functional coordination were handled consistently throughout, contributing to a steady delivery cadence across projects including Masan, Syngenta, P&G, INNOVA US, and GeneSolutions.

Quality Control Reports. Two monthly QC reports were produced for the Masan AI product, each documenting per-image detection metrics, annotation corrections, and error classification across a sample of 1,000 images. These reports were delivered to the Masan client as official quality assurance records and shared with the development team to support model retraining.

Meeting Minutes. Over 30 meeting minutes documents were written throughout the internship, covering client requirement sessions, sprint reviews, and internal alignment meetings. These served as the official written record of decisions and action items agreed upon across all projects.

7.2 Skills Developed

7.2.1 Technical Tools

The internship provided hands-on experience with a broad set of tools, several of which were new at the start of the placement. **Figma** and **FigJam** were used extensively for wireframing and process flow diagramming, developing proficiency in component-based design, auto-layout, and collaborative prototyping. Additionally, UI prototyping using **Stitch AI** and **vibe coding** with React and TypeScript was explored as part of the wireframing workflow, where AI-generated UI components were used as a starting point before refinement in Figma. This introduced a practical understanding of how generative AI tools can accelerate early-stage design work. **Azure DevOps** was used for sprint planning and task tracking, building familiarity with backlog management, sprint board operations, and Agile delivery workflows. **Label Studio** was used for AI image annotation review, providing practical exposure to bounding box labelling and quality control in a computer vision pipeline. **NotebookLM** was adopted for meeting minute drafting, significantly improving documentation speed. **Advanced Excel** skills were applied extensively across requirements tracking, QC reporting, SKU training sheets, and cross-project timeline management.

7.2.2 Professional and Soft Skills

Working across five client projects simultaneously required strong **time management** and the ability to context-switch between different domains, stakeholders, and deliverable formats without losing track of priorities. **Requirements elicitation** skills were developed through repeated cycles of client interviews, written question lists, and iterative document review — learning to translate vague or high-level client requests into structured, developer-ready specifications. **Client communication** improved significantly over the internship, particularly in managing expectations, confirming scope changes, and escalating blockers diplomatically. **Meeting facilitation** was practised in both daily stand-ups and weekly delivery meetings, building confidence in leading structured discussions and keeping sessions on time. The work also strengthened **cross-functional coordination** between BA, development, and QC teams — particularly in translating technical constraints for clients and business requirements for developers. **Handling ambiguous requirements** became a recurring challenge that sharpened the ability to ask the right clarifying questions and document assumptions explicitly.

7.2.3 Domain Knowledge

The internship provided meaningful exposure to several domains with little prior knowledge. Working on the Masan AI product's QC pipeline gave practical insight into **computer vision** concepts including bounding box annotation, object detection metrics (precision, recall, F1 score, IoU), and the iterative model retraining cycle. The INNOVA

US project introduced the emerging field of **agentic AI** — AI systems capable of autonomous, multi-step task execution and self-thinking capabilities. The retail and agriculture projects (Masan, Syngenta, P&G) provided grounding in **FMCG and retail industry** workflows, particularly around SKU management, product recognition, and field data collection. Through the GeneSolutions project, a practical understanding of the full **Software Development Life Cycle (SDLC)** was developed — from initial requirement elicitation and wireframe design through client approval, development handoff, QA, and planned deployment — providing a end-to-end view of how a product feature moves from concept to release within an Agile/Scrum team.

Chapter 8

Lessons Learned and Personal Comments

8.1 Lessons Learned

8.1.1 Client Communication Is Non-Negotiable

One of the most significant lessons from this internship was how much weight client communication carries in real-world BA work. Prior to the placement, it was easy to underestimate this aspect — in academic projects, requirements are usually fixed and given upfront. In practice, clients change their minds, introduce new constraints mid-sprint, and sometimes have difficulty articulating what they actually need. Learning to ask the right clarifying questions, confirm understanding in writing, and manage expectations without creating friction became essential skills developed over time rather than taught all at once. A recurring lesson was never to assume a requirement is understood — if it is not explicitly confirmed, it is not confirmed.

8.1.2 Requirements Are Never Final

Scope changes are not exceptional events — they are the normal state of product development. Throughout the internship, requirements evolved across every project as clients refined their understanding of what they wanted, or as technical constraints became clearer during development. The practical lesson was to design documentation with this volatility in mind: using versioned documents, clearly marking open questions, and building in regular review checkpoints rather than treating any requirements document as a finished artefact.

8.1.3 Documentation Is Harder Than It Looks

Producing clear, complete, and maintainable documentation turned out to be far more demanding than expected. The challenge was not simply writing things down, but structuring information in a way that was useful to multiple audiences — developers who needed precise technical detail, clients who needed readable summaries, and QC teams who needed traceable criteria. Early in the internship, small gaps or ambiguities in documentation caused confusion downstream. Over time, the practice of reviewing documents from the reader's perspective before sharing them — anticipating questions and filling in gaps proactively — significantly improved the quality of deliverables.

8.1.4 Underestimating Complexity Is Costly

Several tasks turned out to be significantly more involved than initially anticipated. UI wireframing was one clear example: what begins as a single user flow quickly expands into dozens of edge-case screens once error states, empty states, permission levels, and responsive variants are considered. Similarly, requirements gathering that appears straightforward often uncovers hidden dependencies once development questions are raised. The lesson learned was to add buffer time when estimating task scope and to break large deliverables into incremental checkpoints rather than attempting to deliver everything at once.

8.1.5 Bridging the Gap Between Roles

Working at the intersection of business, design, and engineering meant constantly translating between different modes of thinking. Developers think in terms of technical feasibility and data structures; clients think in terms of business value and user experience. One of the core skills developed was learning to inhabit both perspectives simultaneously — understanding enough about the technical side to write requirements that developers could act on, while keeping client language accessible and free of jargon. This cross-role awareness is something that classroom learning rarely replicates.

8.2 Personal Comments

Starting the internship was genuinely overwhelming. The transition from university coursework — where tasks are clearly scoped, deadlines are known weeks in advance, and feedback is structured — to a real work environment with live clients and shifting priorities was a significant adjustment. The volume of information, the pace of communication, and the expectation to contribute meaningfully from an early stage were all more demanding than expected. In the first few weeks, there was a noticeable gap between the confidence that academic preparation provides and the uncertainty of navigating real organizational dynamics.

What helped considerably was the team's attitude toward my internship. The senior BA and the leads were aware that the work was demanding for someone at the intern level, and they were consistently willing to answer questions and review my work before it went to clients, and explain the reasoning behind decisions. This made the learning curve steeper in terms of content, but safer in terms of making mistakes — which, in hindsight, was the right environment to learn in.

As the internship progressed and the project load expanded from one or two projects to managing five simultaneously, the pace became noticeably faster. Context-switching between clients with different industries, different tools, and different communication styles was challenging, but it forced the development of a more systematic approach to managing work — using structured tracking sheets, maintaining consistent documentation habits, and learning to break tasks into smaller, manageable pieces rather than attempting to handle everything at once.

On reflection, the experience has shifted my professional interests somewhat. While the BA role confirmed the value of structured thinking, stakeholder management, and documentation discipline, close collaboration with the development team throughout the internship sparked a genuine curiosity about the technical side of product development. The exposure to AI-driven products — particularly computer vision and agentic AI systems — was a particularly interesting dimension that was not anticipated at the start. Moving forward, I have developed a better interest in deepening technical knowledge alongside the business analysis foundation built here.

Chapter 9

Conclusions

List of References